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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Data Analysis Report

Project ID: R26-DS-017

Project Title: Agentic AI for Robust and Socially Aware Autonomous Driving

1. Introduction

1.1. Background:

Autonomous driving systems must make safe and intelligent decisions in complex road environments. However, current systems can struggle with hidden human intent, limited collaboration between vehicles, and socially complex road situations. These issues can lead to unsafe driving, traffic congestion, and accidents. Therefore, this project proposes an Agentic AI framework for robust and socially aware of autonomous driving by combining pedestrian intent prediction, right-of-way negotiation, Theory of Mind-based social reasoning, and collective vehicle learning.

1.2. Research Problem:

Reactive autonomous driving systems often struggle to understand hidden human intent and complex road interactions. They also have limited collaboration between vehicles, which can reduce safety and efficiency. The research problem is to develop an Agentic AI-based autonomous driving framework that can predict road-user intent, support vehicle negotiation, understand social behavior, and share verified knowledge between autonomous vehicles for safer decision-making.

1.3. Objectives:

It Number	Objective	Objective number
IT22245960	To develop an occlusion-aware pedestrian intent prediction model for autonomous driving that can identify pedestrians partially hidden by nearby vehicles and predict their behavior as Cross, Wait, or Hesitate using PIE dataset frames, bounding box information, occlusion context, and movement-related features.	1
IT22265906	To develop a decentralized multi-agent right-of-way negotiation system for autonomous vehicles at unsignalized intersections using V2V-style local awareness, LSTM-based behavior prediction, conflict detection, and SUMO simulation.	2
IT22223012	To develop a Theory-of-Mind-inspired agentic framework for autonomous vehicles that can identify pedestrian mental states from pose, motion, behavioral, and contextual cues, generate proactive	3

	and explainable driving decisions using TTC-based reasoning, and validate safe autonomous vehicle responses using the PIE dataset and SUMO simulation.	
IT22022974	To develop an agentic knowledge sharing framework for autonomous vehicles that can identify known, rare, and new traffic signs, selectively share important knowledge using reinforcement learning, verify shared knowledge globally, and support collective learning between vehicles using the GTSDB dataset and semantic knowledge packages.	4

2. Data Exploration

2.1. Data Collection

IT22245960 – Bandara W A P S

The data for this research was collected from the **PIE dataset**, which contains real-world urban driving videos and pedestrian annotations. The dataset includes video sequences captured from a vehicle-mounted camera, where pedestrians appear in different traffic situations, including visible and occluded conditions. For this research, the PIE videos and annotation files were downloaded and processed. The videos were converted into image frames using OpenCV, and the annotation files were parsed to extract pedestrian details such as frame ID, pedestrian ID, bounding box coordinates, action, crossing status, looking status, and occlusion information. These extracted details were then matched with the corresponding frames to create a labelled dataset for pedestrian intent prediction.

IT22265906 – Nethmini R A I

The main data source used for the "Multi-Agent Negotiation for Right-of-Way in Complex Intersections" component is the inD dataset. This is a real-world traffic trajectory dataset collected from intersection environments using drone-based recordings. For this component, selected recordings from the inD dataset were used to analyze how vehicles move and interact with each other near intersections. The dataset provides important vehicle movement details such as vehicle position, speed, acceleration, direction, frame number, and vehicle class. These data were useful for creating movement-based and interaction-based features needed for vehicle behavior prediction and right-of-way negotiation. The collected data were later preprocessed to create a V2V-style behavior dataset for predicting GO, YIELD, and NEUTRAL vehicle behavior classes.

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The data for this research was collected from the PIE (Pedestrian Intention Estimation) dataset, which contains real-world urban pedestrian interaction videos captured from vehicle-mounted cameras [1]. The video frames and XML annotation files were processed using Python, OpenCV, and MediaPipe to extract pedestrian body landmarks, motion patterns, and contextual information. The extracted data was then used for feature engineering, pedestrian mental-state classification, TTC-based risk estimation, and proactive rule-based autonomous vehicle decision-making.

IT22022974 – Karunaarachchi K A I H

The data for this research was collected from the German Traffic Sign Detection Benchmark (GTSDB) dataset, which contains real-world traffic scene images with annotated traffic signs. The dataset images and annotation files were processed using Python and OpenCV to extract traffic sign class IDs and bounding box coordinates. Cropped sign regions were then used for traffic sign detection, knowledge identification, semantic knowledge package generation, and RL-based selective knowledge sharing.

2.2. Dataset Description:

Adaptive Intent Prediction in Occluded Urban Scenarios:

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Data source	Description	Resource	Size	Key attributes
PIE Dataset	The PIE Dataset was used for pedestrian intent prediction in urban traffic scenes. Videos and annotations were processed to extract frames, match pedestrian bounding boxes, and create Cross, Wait, and Hesitate labels.	PIE videos, annotations, extracted frames, final labelled CSV	50,713 labelled records	frame_id, pedestrian_id, x1, y1, x2, y2, occlusion, action, cross, look, crossing_intent, behavior_label

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Data source	Description	Resource	Size	Key attributes
inD Dataset	Real-world vehicle trajectory dataset collected from intersection scenarios. This dataset was used to study vehicle movement and interaction behavior for right-of-way negotiation. [2]	tracks.csv, tracksMeta.csv, recordingMeta.csv	Selected recordings 07–17. Total tracks.csv: 2,721,779 rows × 17 columns. Total tracksMeta.csv: 3,738 rows × 8 columns. Total recordingMeta.csv: 11 rows × 15 columns.	recordingId, trackId, frame, x, y, xVelocity, yVelocity, xAcceleration, yAcceleration, heading, class

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Proactive Social Compliance Modeling using Agentic Theory of Mind:
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Data source	Description	Resource	Size	Key attributes
PIE Dataset	PIE Dataset containing pedestrian intention, crossing behavior, bounding box annotations, and urban traffic interaction videos captured from vehicle-mounted cameras [1].	PIE videos, XML annotations, extracted frames, pose landmarks, processed feature .npy files, sequence .npy files	53 videos, 300K+ frames, 1800+ pedestrian tracks, 36 extracted features	frame_id, pedestrian_id, x1, y1, x2, y2, body landmarks, current_speed, acceleration, head_angle, forward_lean, hesitation_cycles, looks_left, looks_right, vehicle_distance, traffic_light_state, pedestrian_density, mental_state_label

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Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing:
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Data source	Description	Resource	Size	Key attributes
GTSDB Dataset	German Traffic Sign Detection Benchmark dataset used for traffic sign detection and knowledge identification	Traffic scene images, gt.txt annotations, cropped sign images	600 training images, 300 evaluation images, 43 traffic sign classes	image_id, class_id, bounding boxes, sign category, confidence score, novelty value

2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

Adaptive Intent Prediction in Occluded Urban Scenarios:
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	1	2	3	4
Data source 1: PIE Dataset	Relevant to urban pedestrian intent prediction.	Provides frames, annotations, bbox, action, and occlusion data.	Support Cross, Wait, and Hesitate Label Creation.	Suitable for C1 occlusion-aware model development.

Multi-Agent Negotiation for Right-of-Way in Complex Intersections:
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	1	2	3	4
Data source 1: inD Dataset	Relevant because it provides real vehicle movement data such as position, speed, acceleration, and heading. [2] [3]	Relevant because the data can be converted into time sequences for LSTM training. [4]	Relevant because nearby vehicle position and movement can be used to identify interaction and conflict situations.	Relevant because it helps create realistic vehicle behavior for testing in simulation.

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	1	2	3	4
Data source: PIE Dataset	Relevant to pedestrian pose, motion, and behavioral feature extraction	Provides pedestrian videos, annotations, and temporal interaction data for LSTM and GRU mental-state classification	Supports TTC-based risk estimation and proactive rule-based AV decision-making	Suitable for SUMO simulation validation and proactive autonomous vehicle behavior analysis

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Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing:
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	1	2	3	4
Data source: GTSDB Dataset	Relevant to traffic sign detection and classification	Provides traffic sign images, annotations, and bounding box data	Supports knowledge identification and semantic package generation	Suitable for RL-based knowledge sharing and collective learning experiments

3. Methodology

3.1. Data Preprocessing:

Objective 1:

Adaptive Intent Prediction in Occluded Urban Scenarios:

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	Transformation technique							
	Data Cleaning	Data Filtering	Feature Engineering	Date/Time Transformation	Data Mapping	Data Scaling	Data Encoding	Sequence Creation
Data source:	Removed missing image paths, duplicate frames, invalid annotations	Filtered partially occluded pedestrians and relevant urban scenes	Extracted bounding box size, center coordinates, occlusion status, pedestrian motion features	Converted frame numbers into temporal sequences	Mapped annotations to behavior labels (Cross, Wait, Hesitate)	Normalized bounding box coordinates and numerical features	Label Encoding for behavior classes and occlusion categories	Created sequential frame windows for temporal intention prediction

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Objective 2:
Multi-Agent Negotiation for Right-of-Way in Complex Intersections:

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	Transformation technique							
	Data Cleaning	Data Filtering	Feature Engineering	Date/Time Transformation	Data Mapping	Data Scaling	Data Encoding	Sequence Creation
Data source: inD Dataset	Removed missing or invalid vehicle records.	Selected vehicle-related data and removed unnecessary road users.	Calculated speed, acceleration, heading, distance to intersection, and ETA.	Converted frame numbers into time-based values.	Mapped vehicle movement into approach, exit, and zone values.	Scaled numerical features before training.	Converted behavior labels into numerical format.	Created 10-step vehicle movement sequences for LSTM input.

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Objective 3:
Proactive Social Compliance Modeling using Agentic Theory of Mind:
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	Transformation technique							
	Data Cleaning	Data Filtering	Handling Missing Data	Feature Engineering	Data Mapping	Outlier Detection and Handling	Data Standardization	Sequence Generation
Data source: PIE Data set	Removed invalid pedestrian detections, missing annotations, empty landmark frames, and fully occluded pedestrian samples.	Selected valid pedestrian tracks with sufficient frame sequences for temporal modeling.	Applied default fallback values for unavailable contextual attributes and missing feature values.	Computed 36 motion, pose, behavioral, and contextual features from pedestrian tracks and pose landmarks.	Mapped PIE annotations and behavioral cues into mental-state classes.	Replaced nan and infinite numerical values using safe numerical conversion methods.	Applied Standard Scaler normalization before LSTM and GRU model training.	Generated overlapping 30-frame temporal sequences for deep learning-based temporal classification.

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Objective 4:
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	Transformation technique							
	Data Cleaning	Data Filtering	Feature Engineering	Data Mapping	Data Scaling	Data Encoding	Cropped Image Generation	Knowledge Package Generation
Data source: GTSDB Dataset	Removed invalid or duplicate annotations.	Selected traffic sign classes and removed unrelated data.	Calculated confidence score, novelty value, and importance score.	Mapped traffic signs into prohibitory, danger, and mandatory categories.	Normalized image pixel values before model training.	Converted class labels into YOLO numerical format.	Cropped detected traffic sign regions from road images.	Generated semantic knowledge packages with metadata and embeddings

3.2. Scalability

Adaptive Intent Prediction in Occluded Urban Scenarios:
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PIE Dataset Designed preprocessing and sequence generation pipeline to handle large-scale pedestrian video data efficiently for future dataset expansion. The preprocessing and feature extraction pipeline was designed with scalability in mind, allowing efficient handling of large pedestrian trajectory datasets and future integration of additional sensors, occlusion contexts, and urban traffic scenarios. The modular architecture supports expansion to larger datasets and real-time autonomous driving applications.

Multi-Agent Negotiation for Right-of-Way in Complex Intersections:

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The inD dataset is suitable for this component because it contains real-world vehicle trajectory data from intersection scenarios. [2] [3] After preprocessing, the created V2V behavior dataset contains enough records to train and test the LSTM-based behavior prediction model. [4] The dataset can also be scaled by adding more inD recordings, more intersection scenarios, and more vehicle interaction cases. This makes it suitable for improving the right-of-way negotiation system in the future.

Proactive Social Compliance Modeling using Agentic Theory of Mind:
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The PIE dataset is suitable for this component because it contains real-world pedestrian interaction videos captured from vehicle-mounted cameras [5]. After preprocessing, the created feature dataset contains 36 motions, pose, behavioral, and contextual features, which are sufficient to train and test the LSTM and GRU-based pedestrian mental-state prediction models [6] [6]. The dataset can also be scaled by adding more pedestrian videos, more urban crossing scenarios, and more culturally diverse pedestrian behavior samples. This makes it suitable for improving proactive AV decision-making in the future.

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing:

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The GTSDB dataset is suitable for this component because it contains real-world traffic sign images with annotated bounding boxes and multiple traffic sign categories. After preprocessing, the generated traffic sign knowledge dataset contains sufficient samples to train and evaluate the YOLOv8n-based traffic sign detection and knowledge identification models. The dataset can also be scaled by adding more traffic sign images, additional road environments, and newly verified knowledge packages collected from multiple vehicles. This makes it suitable for improving collective learning and selective knowledge sharing in autonomous vehicle systems in the future [7] [8].

3.3. Feature extraction

Adaptive Intent Prediction in Occluded Urban Scenarios:
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Important features were extracted from the PIE pedestrian dataset to support pedestrian intention prediction in partially occluded urban traffic scenarios.
The extracted features include:

Pedestrian movement features:

pedestrian bounding box coordinates, center position, movement direction, walking speed, frame-to-frame displacement, and motion patterns.

Occlusion-related features:

occlusion status, occlusion type (none, partial, full), visibility level, hidden body area, nearby vehicle presence, and pedestrian location relative to the occluding object.

Contextual traffic features:

crossing intent, looking behavior, pedestrian gesture information, scene context, and spatial position within the road environment.

Temporal sequence features:

sequential frame observations, temporal movement changes, historical pedestrian behavior patterns, and short-term motion trends.

Target label:

Pedestrian behavior was labelled as CROSS, WAIT, or HESITATE.

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Multi-Agent Negotiation for Right-of-Way in Complex Intersections:

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Important features were extracted from the inD vehicle trajectory data to support behaviour prediction and right-of-way negotiation.
The extracted features include:

Vehicle movement features:
vehicle position, speed, acceleration, heading, and movement direction.

Intersection-related features:
distance to intersection, estimated time of arrival, approach direction, exit direction, and vehicle zone.

V2V-style interaction features:
nearest vehicle distance, relative speed, minimum ETA gap, conflict count, right-side vehicle presence, right-side vehicle ETA, and right-side vehicle distance.

Target label:
vehicle behavior was labelled as GO, YIELD, or NEUTRAL.

Proactive Social Compliance Modeling using Agentic Theory of Mind:
IT22223012 – Withanage W W D T H

Important features were extracted from the PIE pedestrian interaction dataset to support pedestrian mental-state prediction and proactive autonomous vehicle decision-making.
The extracted features include:

Motion features:
pedestrian speed, average speed, acceleration, deceleration, speed variance, step frequency, and movement direction.

Pose-related features:
head angle, head turn frequency, body orientation, forward lean, lateral lean, shoulder angle, hip angle, and foot angles extracted using MediaPipe pose landmarks [9] .

Behavioral features:
hesitation cycles, pause duration, looks left/right behavior, direction changes, movement probability, and distance change rate.

Contextual features extracted from annotations:
traffic light state, crosswalk presence, pedestrian action, crossing intention, occlusion level, road width, vehicle distance, and pedestrian density.

Target label:
pedestrian behavior was mapped into six mental-state classes: Waiting, Hesitant, Committed, Distracted, Aggressive, and Jaywalk.

The selected pose, motion, and contextual features are supported by previous pedestrian intention prediction studies that used body pose, temporal movement patterns, and contextual reasoning for behavior understanding [5] [6] [10].

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing:
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Important features were extracted from the GTSDB traffic sign dataset and detection pipeline outputs to support traffic sign detection, novelty identification, and selective knowledge sharing.
The extracted features include:

Traffic sign detection features:
bounding box coordinates, detection confidence, cropped sign region, and traffic sign category.

Visual features:
image width, image height, bounding box area, image area, sign shape, sign color, and visual prominence.

Novelty-related features:
novelty value, known/rare/new classification, confidence threshold values, and importance score [12].

Knowledge package features:
embedding vectors, package ID, vehicle ID, timestamp, verification status, and global confidence score.

RL-based sharing features:
importance score, novelty value, visual prominence, sign shape, sign color, and SHARE/IGNORE action labels.

Target label:
knowledge sharing behaviour was labelled as SHARE or IGNORE for RL-based selective knowledge sharing [13].

4. Modelling and Results

Adaptive Intent Prediction in Occluded Urban Scenarios:

IT22245960 – Bandara W A P S

For this research, different models were developed to predict pedestrian intent in occluded urban driving scenarios. The main objective was to classify pedestrian behavior into three classes: **Cross**, **Wait**, and **Hesitate**.

First, a **ResNet18 baseline model** was trained using image frames from the prepared PIE dataset. This model used only image-based features to classify pedestrian behavior. Since the dataset was imbalanced, especially with fewer Cross samples compared to Wait samples, class weighting was applied during training to reduce bias toward the majority class.

After the baseline model, a **C1 Occlusion-Aware Model** was developed. This model uses two types of inputs: image features extracted using ResNet18 and context features such as occlusion status, occlusion type, bounding box size, pedestrian position, and looking status. These features were combined using a fusion layer to improve prediction performance in occluded scenarios.

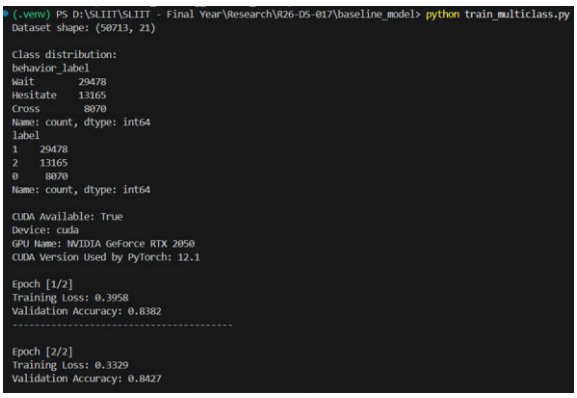
The baseline model achieved approximately **83% accuracy**, while the C1 Occlusion-Aware Model improved the accuracy to around **86%**. The C1 model also showed better performance in occluded cases, especially for vehicle-related occlusion scenarios. This shows that adding occlusion and context features helps the model make better intent predictions for partially visible pedestrians.

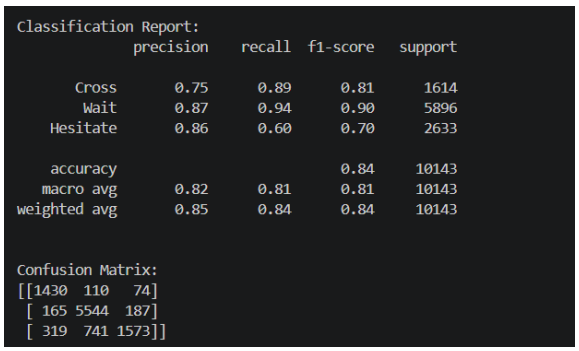
A temporal movement-based model was also tested using features such as `dx`, `dy`, `speed`, `bbox_area_change`, and rolling average speed. This helped analyze pedestrian movement across frames and supported intent prediction based on motion patterns.

4.1. Key Insights:

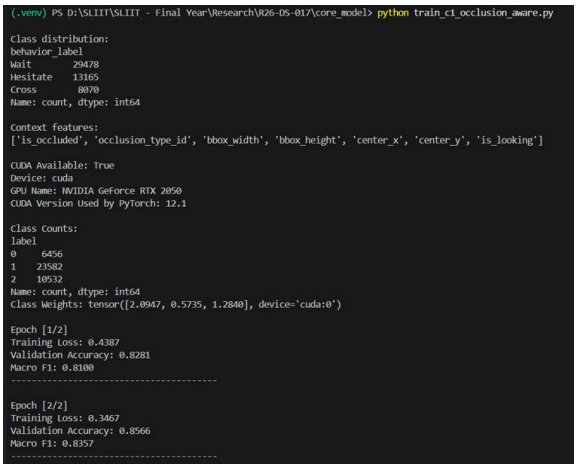
- The dataset has a class imbalance. **Wait** has the highest number of samples, while **Cross** has fewer samples.
- Occlusion is important for this research because pedestrians can be partially or fully hidden by vehicles.
- Bounding box features such as `x1`, `y1`, `x2`, `y2`, width, height, and position help identify pedestrian location and visibility.
- Movement features such as `dx`, `dy`, `speed`, and `avg_speed_5` help understand whether a pedestrian is crossing, waiting, or hesitating.
- The **C1 Occlusion-Aware Model** performed better than the baseline model because it used both image features and occlusion context.
- Vehicle occlusion analysis helped focus on pedestrians hidden by nearby/front vehicles.

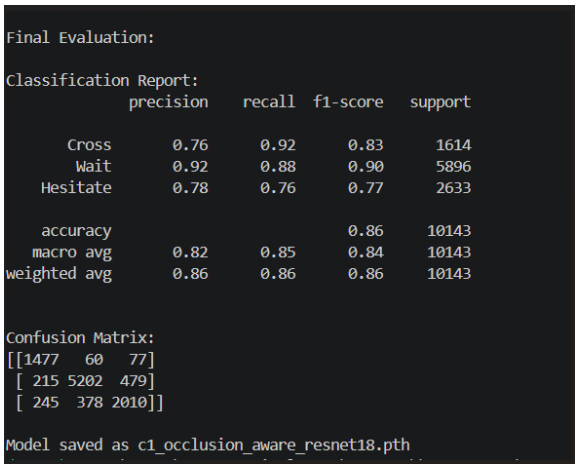
Baseline Model Results





Core Model Results





Challenges Faced During Data Analysis:

- The dataset had a class imbalance. The **Wait** class had more samples than **Cross** and **Hesitate**.
- Some pedestrian frames were partially or fully occluded, making intent prediction more difficult.
- Matching extracted video frames with annotation data was challenging because frame IDs and image paths had to be checked carefully.
- Some annotations had missing or unclear values, such as undefined gestures or limited occlusion details.
- Vehicle-hidden pedestrian scenarios were limited in the PIE dataset, so simulator validation may be needed later.
- Detecting only partial pedestrian body parts was difficult because they are not always clearly visible in the frame.

Multi-Agent Negotiation for Right-of-Way in Complex Intersections:
IT22265906 – Nethmini R A I

For this component, an LSTM-based vehicle behavior prediction model was developed using the processed dataset created from the inD vehicle trajectory dataset. [4] The model was trained to predict vehicle behavior into three classes: GO, YIELD, and NEUTRAL. These predictions support the right-of-way negotiation process by helping the system understand whether a vehicle is likely to move, slow down, or stay neutral before reaching the intersection.

The model was evaluated using accuracy, balanced accuracy, macro F1-score, classification report, and confusion matrix. These evaluation methods were used to check whether the model performs well for all behavior classes, not only the majority class.

4.2. Key Insights:

The processed V2V behavior dataset was created from selected inD dataset recordings [2] [3]. The final dataset included vehicle movement and interaction features such as speed, acceleration, heading, distance to intersection, ETA, nearest vehicle distance, ETA gap, conflict count, and right-side vehicle information.

The final dataset contained:

Total records: 95,326
Total features: 30
Classes: GO, YIELD, NEUTRAL

The label distribution was:

NEUTRAL: 58,007 samples
GO: 24,732 samples
YIELD: 12,587 samples

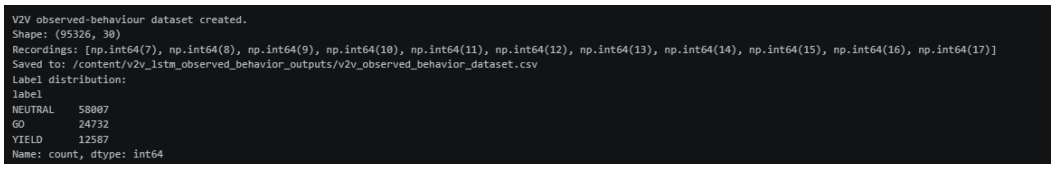


Figure 1 - Dataset Details

This shows that most vehicles were in a neutral movement state, while GO and YIELD behaviours were lower compared to NEUTRAL

The LSTM model was trained using 10-step vehicle movement sequences. This helped the model learn behaviour patterns over time instead of using only one single frame. The final model achieved:

Normal test accuracy: 93.89%
Balanced test accuracy: 93.35%
Macro F1-score: 89.50%

These results show that the model performed well in predicting vehicle behaviour. The confusion matrix and classification report showed that the model predicted the NEUTRAL class very well, while GO and YIELD were also predicted with acceptable performance. This means the selected movement and V2V-style interaction features are useful for vehicle behavior prediction at intersections.

```
Test loss: 0.17890815436840057
Normal test accuracy: 0.9389221668243408
Balanced test accuracy: 0.9335106058142677
Macro F1 score: 0.8950270189520807
Classification report:
              precision    recall  f1-score   support

          GO       0.83      0.95      0.88      2895
     NEUTRAL       0.99      0.94      0.96     13906
       YIELD       0.77      0.91      0.84      1569

    accuracy                           0.94     18370
   macro avg       0.86      0.93      0.90     18370
weighted avg       0.95      0.94      0.94     18370

Confusion matrix:
           GO  NEUTRAL  YIELD
GO       2746       73     76
NEUTRAL   493    13071    342
YIELD      84       54   1431
Prediction sample head:
   true_label predicted_label
0          GO              GO
1          GO              GO
2          GO              GO
...
16      YIELD           YIELD
17      YIELD           YIELD
18      YIELD           YIELD
19         GO           YIELD
```

Figure 2- Accuracy Details

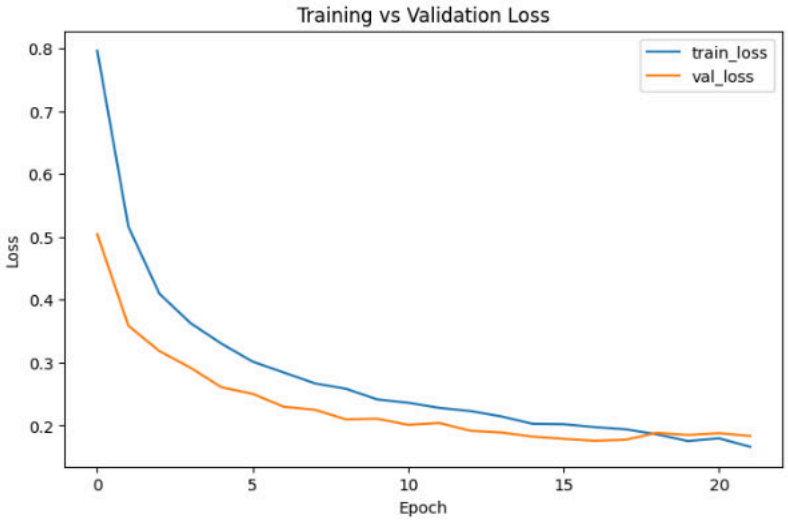


Figure 3 - Training vs Validation Loss Graph

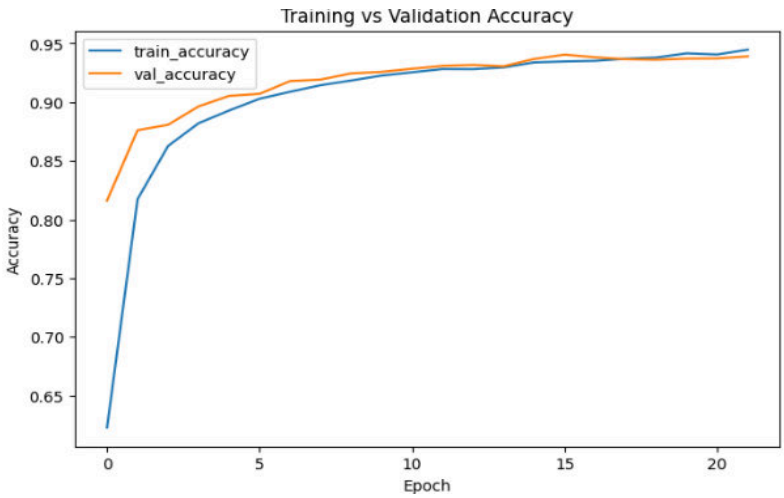


Figure 4 - Training Vs Validation Accuracy Graph

4.3. Challenges Faced During Data Analysis:

Several challenges were faced during the data analysis stage. The raw inD dataset did not directly include all required features for this component. Features such as

ETA, nearest vehicle distance, ETA gap, conflict count, right-side vehicle presence, and right-side vehicle distance had to be manually calculated using vehicle trajectory data.

Another challenge was that the dataset did not directly include behaviour labels such as GO, YIELD, and NEUTRAL. Therefore, labels had to be created using vehicle movement patterns and interaction logic.

Class imbalance was also a challenge because the NEUTRAL class had more samples than the GO and YIELD classes. Because of this, normal accuracy alone was not enough to evaluate the model. Balanced accuracy, macro F1-score, classification report, and confusion matrix were used to check the model performance more fairly.

Another challenge was avoiding data leakage. Future information and target-related columns had to be removed before training, so the model would learn from real input features only.

Overall, the data analysis stage required careful feature engineering, label creation, sequence preparation, and model evaluation to make the behavior prediction model reliable.

Proactive Social Compliance Modeling using Agentic Theory of Mind:
IT22223012 – Withanage W W D T H

The results section presents the key findings obtained from analyzing the processed PIE pedestrian interaction dataset using motion, pose, behavioral, and contextual features. The analysis was conducted using LSTM, GRU, and ensemble-based temporal deep learning models together with TTC-based proactive AV decision reasoning [5] [11] . The results evaluate how effectively the extracted pedestrian features support mental-state prediction and proactive autonomous vehicle decision-making.

4.4. Key Insights:

The processed pedestrian behavior dataset was created from PIE videos, XML annotations, extracted pose landmarks, and engineered feature values. The final feature dataset contained 73,743 frame-level records with 36 features per frame.

The frame-level class distribution was:

Class	Count	Percentage
Waiting	33,069	44.8%
Hesitant	4,479	6.1%
Committed	13,064	17.7%
Distracted	5,358	7.3%
Aggressive	11,745	15.9%
Jaywalk	6,028	8.2%

After sequence generation using 30 consecutive frames per sequence, the final model dataset contained:

Dataset	Shape
Training	2,721 × 30 × 36
Validation	583 × 30 × 36
Testing	584 × 30 × 36

A track-aware split was used, and the leakage check showed **0 overlap** between train, validation, and test pedestrian tracks. This means the same pedestrian track was not shared across different splits.

The LSTM model achieved:

Test accuracy - 77.1%
Macro F1-score - 69.7%
Weighted F1-score - 77.2%
Best validation accuracy-79.8%

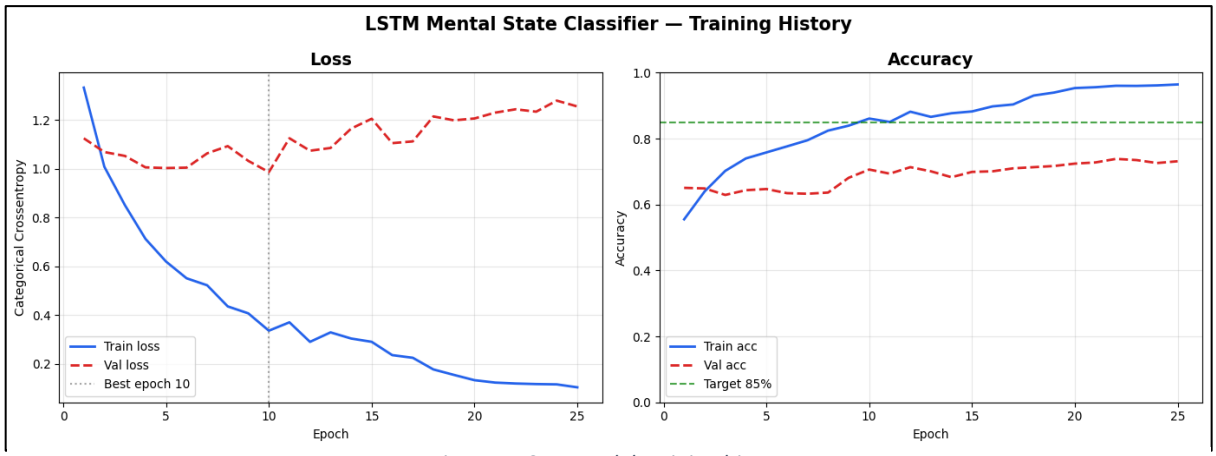


Figure 5:LSTM model training history

The GRU model achieved:

Test accuracy - 74.7%

Macro F1-score - 68.5%

Weighted F1-score - 75.4%

Best validation accuracy - 78.7%

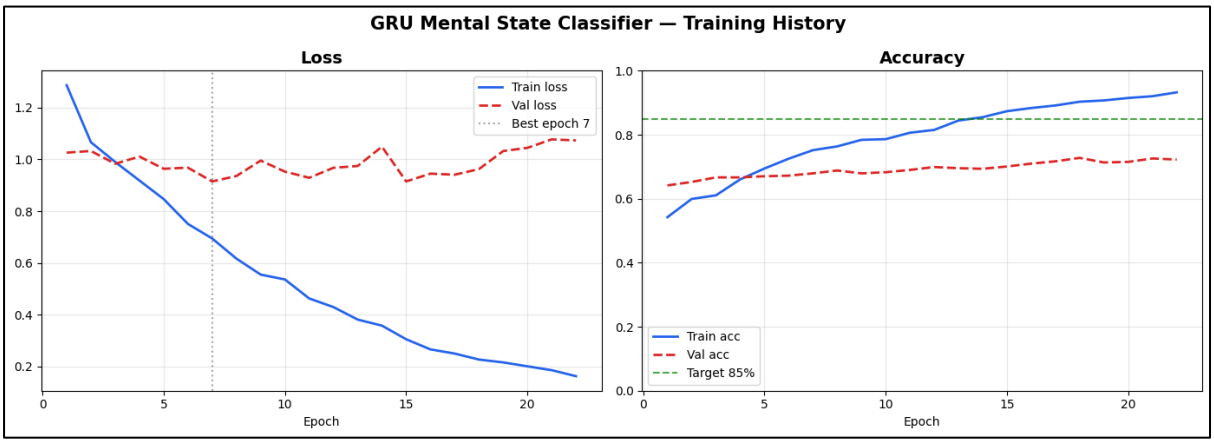


Figure 6:GRU model training history

The LSTM+GRU ensemble gave the best overall result:

Model	Accuracy	Weighted F1-score
LSTM	77.05%	77.19%
GRU	74.66%	75.44%
Ensemble	78.08%	78.37%

This shows that combining LSTM and GRU predictions improved performance compared to using a single model [11].

The ensemble classification report showed stronger performance for Waiting, Committed, and Aggressive classes. However, Hesitant, Distracted, and Jaywalk had lower performance because their movement patterns were more similar and harder to separate.

The ensemble model achieved:

- Waiting - F1-score: 0.81
- Hesitant - F1-score: 0.55
- Committed - F1-score: 0.86
- Distracted - F1-score: 0.70
- Aggressive - F1-score: 0.85
- Jaywalk - F1-score: 0.58

```
==========================================
ENSEMBLE — PER CLASS REPORT
==========================================
              precision    recall  f1-score

     Waiting       0.86      0.76      0.81
     Hesitant      0.48      0.64      0.55
   Committed       0.83      0.89      0.86
  Distracted       0.60      0.84      0.70
  Aggressive       0.85      0.85      0.85
     Jaywalk       0.59      0.57      0.58

    accuracy                           0.78
   macro avg       0.70      0.76      0.72
weighted avg       0.79      0.78      0.78
```

Figure 7:ensemble model report

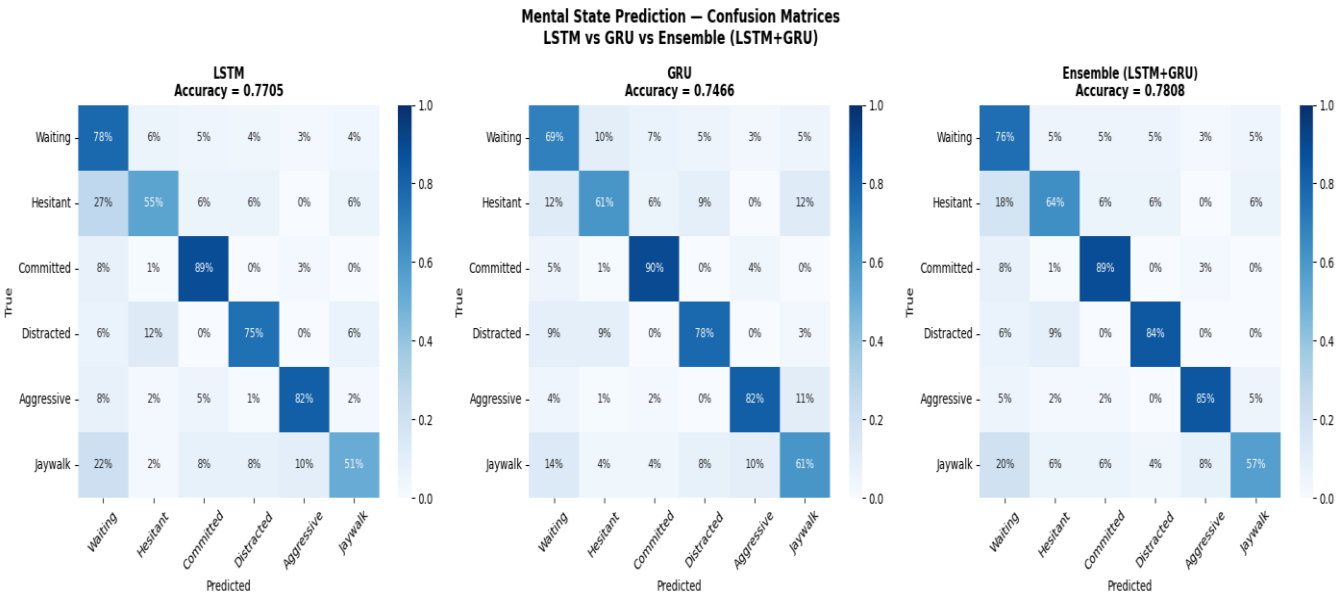


Figure 8:Confusion Matrices

4.5. Challenges Faced During Data Analysis:

Several challenges were identified during feature extraction, preprocessing, model training, and evaluation.

The dataset was imbalanced. Waiting had the highest number of samples, while Hesitant and Distracted had much fewer samples. This made minority classes harder to predict accurately.

Some pedestrian tracks did not contain enough frames to create 30-frame temporal sequences. These short tracks had to be removed during sequence generation.

Pose landmark extraction was affected by occlusion, small pedestrian size, and low-quality frames. This created unstable or missing landmark values in some cases [9].

The pipeline required missing-value and invalid-value handling. The notebook checked for NaN and infinite values and used safe numerical conversion before saving the final feature arrays.

Feature extraction was computationally expensive because many frames had to be processed with pose estimation and annotation matching.

The LSTM and GRU models showed good learning performance, but the final accuracy did not reach the original target of 85%. The final ensemble accuracy was 78.08%, so further improvement is needed.

Federated Collective Learning Between Autonomous Vehicles using Agentic Knowledge Sharing:
IT22022974 – Karunaarachchi K A I H

For this component, a YOLOv8n-based traffic sign detection and knowledge identification system was developed using the processed GTSDB dataset. The system was designed to detect traffic signs, identify known, rare, and new signs, and support RL-based selective knowledge sharing between autonomous vehicles. The generated semantic knowledge packages and verification pipeline were used to support collaborative learning and intelligent knowledge broadcasting between vehicles [8] [12].

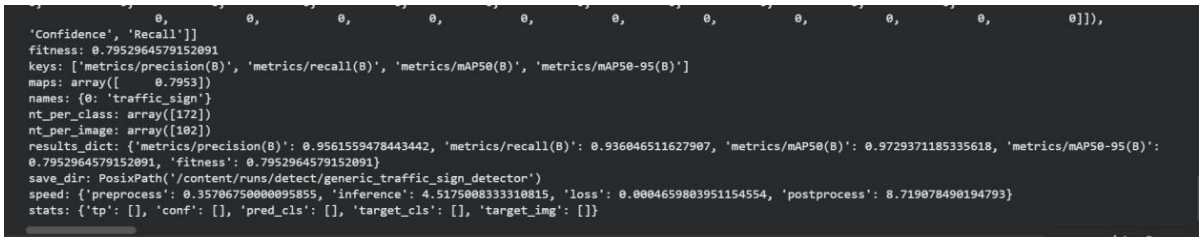
The detection and RL-based sharing models were evaluated using precision, recall, mAP, accuracy, F1-score, classification report, and confusion matrix. These evaluation methods were used to measure traffic sign detection performance, selective knowledge sharing behaviour, and verification reliability [7] [13].

4.6. Key Insights:

The processed traffic sign dataset was created from the GTSDB dataset images, annotations, cropped sign regions, and generated semantic knowledge packages. The final dataset included traffic sign features such as bounding box coordinates, confidence scores, novelty values, visual prominence, sign shape, sign colour, and importance scores.

The traffic sign detection model achieved:

Metric	Result
Precision	95.61%
Recall	93.60%
mAP@50	97.29%
mAP@50-95	79.52%



These results show that the YOLOv8n model successfully detected traffic signs from road scene images with high detection accuracy [8] [7].

The knowledge identification module successfully classified traffic signs into KNOWN, RARE, and NEW categories using confidence threshold logic.

The threshold distribution used was:

Knowledge Type	Threshold
KNOWN	≥ 0.7
RARE	0.3 – 0.7
NEW	≤ 0.3

The RL-based selective knowledge sharing model was trained to predict whether detected knowledge should be SHARE or IGNORE. The final RL model achieved:

Metric	Result
Accuracy	69.07%
Recall	100%
F1-Score	81.70%

```
============================================================
Accuracy:  0.6907 (69.07%)
Precision: 0.6907
Recall:    1.0000
F1-Score:  0.8170

Model predictions:
  Predicted SHARE: 236
  Predicted IGNORE: 0

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```

The results show that the RL model successfully learned selective knowledge sharing behaviour and reduced unnecessary knowledge propagation between vehicles [12]. However, the model still requires more balanced SHARE and IGNORE examples for improved decision-making performance.

The semantic knowledge package generation and global verification modules successfully verified knowledge before broadcasting, supporting collaborative learning between autonomous vehicles [13].

4.7. Challenges Faced During Data Analysis:

Several challenges were faced during the data analysis stage. The GTSDB dataset did not directly contain novelty-related labels such as KNOWN, RARE, and NEW. Therefore, these labels had to be generated using confidence threshold logic and knowledge identification rules [12].

Another challenge was that the RL training dataset contained more SHARE examples compared to IGNORE examples. This caused data imbalance during RL model training and affected selective sharing performance [13].

Small traffic signs and low-prominence signs were also difficult to detect accurately under some road conditions. Because of this, visual prominence and novelty weighting strategies were introduced during importance score calculation [8] [12].

Another challenge was catastrophic forgetting during incremental model updates. Initially, full fine-tuning caused the model to forget previously learned traffic sign knowledge. Therefore, parameter freezing and low learning rate incremental tuning approaches are currently being tested to improve stable model updating [7].

Overall, the data analysis stage required careful feature engineering, novelty identification, semantic package generation, RL training, and verification filtering to support reliable collective learning between autonomous vehicles [8] [12] [7].

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